



## Effects of Spiritual Group Therapy on Caregiver Strain in Home Caregivers of the Elderly with Alzheimer's Disease☆



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### ABSTRACT

**INTRODUCTION:** Care of patients with Alzheimer's disease is one of the most difficult types of care that exposes the caregiver to a high level of care strain. The present research aimed at determining the effect of spiritual care on caregiver strain of the elderly with Alzheimer's disease.

**METHODS:** An experimental study was carried out on 100 caregivers who were selected by convenience sampling and randomly divided into intervention, control one and control two groups. Group spiritual therapy was performed on the intervention group for five weeks, Control one participate in the group sessions without any particular interventions, and control two received no interventions. Data was collected through a demographic questionnaire and Robinson's (1983) Caregiver Strain Index, and analyzed using the Chi-square, Fisher's Exact test, one-way analysis of variance and paired *t*-test. Statistical significance level was considered as 0.05.

**RESULTS:** In the intervention group mean of the posttest care strain score  $32.43 \pm 2.73$  was significantly lower than pretest  $37.16 \pm 1.26$  ( $P < 0.001$ ). The mean posttest score of care strain was significantly lower in the intervention group compared to the two other groups ( $P < 0.001$ ).

**CONCLUSION:** Spiritual care can reduce care strain in home caregivers of the elderly with Alzheimer's disease.

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The world's population is aging (United-Nations, 2015), and Aging is the greatest risk factor for Alzheimer's disease (Galvan & Hart, 2016). Late onset Alzheimer's disease is age dependent and occurs in  $\geq 65$  years, accounts for  $>95\%$  of affected patients (Reitz & Mayeux, 2014). After the age 65, Alzheimer prevalence doubles each five years and 40–50% of the elderly  $>85$  years have Alzheimer's disease (Yektatalab, Kaveh, Sharif, Fallahi-Khoshknab, & Petramfar, 2012; Zohari, Khatouni, Abed, Alavi, & Yaghmaei, 2006). Global epidemic of Alzheimer's disease was confirmed by demographic studies (Alzheimer's Association, 2011). Alzheimer patients are dependent to caregivers to meet their needs due to cognitive, functional, and behavioral impairments (Lenderking et al., 2013), and informal caregivers which generally were the patient's family members provide the majority of care (Danucalov et al., 2013; Wennberg, Dye, Streetman-Loy, & Pham, 2015).

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Seven out of every 10 patients with Alzheimer's disease are cared at home (Mohammadi & Babaie, 2011). Taking care of these patients is considered a 36-hour day job, which affects the caregivers' life in different aspects, including health, employment and income (Bastani, Ghasemi, Negarandeh, & Haghani, 2012; Fallahi-Khoshknab, Sheikhona, Rahgouy, Rahgozar, & Sodagari, 2014). Caregivers of the elderly believe their tasks cause mental fatigue, sense of inability, care strain, psychological disorders, concerns and fears (Nikraves, 2008). Care strain is the amount of physical, mental and psychosocial stress and efforts that caregivers have to bear when providing care and meeting the patient's needs (Chou, Liu, & Chu, 2002). The strain of taking care of the elderly with Alzheimer's disease is associated with numbers of adverse consequences, such as depression, high psychotropic drug use, reduced physical health, impaired quality of life and immune system deficiency (Bruvik, Ulstein, Ranhoff, & Engedal, 2012). In Iran, care strain might be overwhelming due to the strong affections in families which were not allowing them to leave their elderly in nursing homes. In Iranian culture, informal home care is an important duty of family members (Abdollahpour, Noroozian, Nedjat, & Majdzadeh, 2012).

Several interventions which aimed to relieve care strain have been developed. Strategies include education interventions, which utilize caregivers by problem solving skills; support group interventions, which provide support, normalize the experience and improve social relations; psychoeducation interventions, counseling interventions,

which develop plans to meet individual needs and respite care interventions (Van't Leven et al., 2013; Wennberg et al., 2015).

Despite the growing elderly population in Iran, no plans have been devised for providing formal care to them, especially in the long term care (Mohammadi & Babaie, 2011). Although moderating to high level of care strain observed in >50% of caregivers, Care strain imposed on the caregivers has not been sufficiently addressed. It is an issue which needs more attention from the national healthcare policy makers (Lotfi-Kashani, Vaziri, Arjmand, Mousavi, & Hashemieh, 2012). If the problems of caregivers remain unsolved, this crucial source of care will get physically and mentally exhausted, they will lose their efficacy in providing care, and it will lead adverse consequences for Alzheimer patients (Yektatalab et al., 2012; Zohari et al., 2006). By considering the cultural, social and belief differences between Iranian and people of other countries, which may affect the type of spiritual care provided and its outcome (Napoles, Chadiha, Eversley, & Moreno-John, 2010; Spurlock, 2005) the present experimental study was conducted to fill the existing gaps in Iran and to investigate the effect of spiritual care on care strain in home caregivers of patients with Alzheimer's disease with the study hypothesis that spiritual care reduces care strain in home caregivers of the elderly with Alzheimer's disease.

Spirituality improves mental health, well-being, physical health and health behaviors, decreases depression, anxiety and substance abuse (Koenig, 2014). Spiritual care ameliorates quality of life in family caregivers of the elderly with Alzheimer's disease (Mahdavi, Fallahi-Khoshknab, Mohammadi, & Hosseini, 2015). Spiritual care is a complementary method that can help the caregiver for providing better care and reduces the care strain (Breiddal, 2011). Spirituality is an intrinsic propensity to create meanings through a sense of attachment to the transcendent level of life that leads to achieve the meaning and purpose of life (Rahnama, Fallahi-Khoshknab, Maddah, & Ahmadi, 2012). Spiritual care identifies and responds to people's spiritual and psychological needs in face of trauma, disease or concern. It can involve the need for meaning and purpose, self-value and self-respect, self-expression and self-interpretation, protection of faith or perhaps the need for religious practices and rituals and prayer (Fallahi-Khoshknab & Mazaheri, 2008). In nursing, spiritual intervention is a unique dimension of care and cannot be replaced with psychological or social care. It responses to fundamental human questions such as the meaning of life, pain, suffering and death (Bamdad, Fallahi-Khoshknab, Dalvandi, & Khodayi-Ardakani, 2013). Spiritual care interventions have the aim of facilitating the spiritual growth and ultimately promote spiritual health, resulting in existential uniformity and integrity (Hammer, Mogensen, & Hall, 2009; Lin, Liou, & Chen, 2008).

Spiritual care is known as an intervention which affects care strain in caregivers of patients with Alzheimer (Spurlock, 2005) and other diseases (Jones, 2012; Pallangyo & Mayers, 2009; Tan et al., 2012). Tan et al. (2012) showed that spiritual care is significantly related to care strain in caregivers of schizophrenia patients, and it can help them in coping with the problems which is made by taking care of these patients (Tan et al., 2012). A study conducted by Pallangyo and Mayers (2009) showed that spiritual care can help caregivers of women living with HIV<sup>1</sup> to restore the circumstances, regain life control and mental adaptation and is known as an important coping mechanism (Pallangyo & Mayers, 2009). Jones (2012) study illustrated that religion and spirituality can affect the care role and experience of families with cancer children (Jones, 2012). The results of a study conducted by Spurlock (2005) showed that spiritual care is an important coping mechanism among female caregivers of patients with Alzheimer's disease (Spurlock, 2005). Rahmati, Fallahi-Khoshknab, Rahgouy and Rahgozar (2010) study demonstrated that spiritual-religious care is effective in improving the mental status of patients with schizophrenia (Rahmati et al., 2010). In another study, Lotfi-Kashani et al. (2012) represented

that spiritual care can reduce distress and anxiety in mothers of children with cancer (Lotfi-Kashani et al., 2012). A study conducted by Sharifi and Fatehizade (2012) confirmed the effectiveness of the spiritual-religious coping role in the mental health of home caregivers (Sharifi & Fatehizade, 2012).

## METHOD

### PARTICIPANTS

The statistical population was 4000 main family caregivers of the elderly with Alzheimer's disease registered in Iran Alzheimer's Association in 2014. Sampling was performed through the convenience sampling method. The main caregivers who met the inclusion criteria which consisting of participation interest, the ability to read and write, the ability to communicate verbally in Persian, being a Muslim, having a minimum of one year care experience, being the main caregiver of the elderly by self-report and not suffering from any psychological disorders or using psychotropic drugs according to self-reports entered the study after taking the informed consent. Study exclusion criteria consisted of being absent for more than two sessions, participant's withdrawal from participation in the spiritual care interventions or the death of the elderly under care during the study.

According to the result of a pilot study on eight cases, the expected difference before and after intervention was  $1.75 (37 \pm 2.2 \text{ vs } 35.25 \pm 2.1)$  and statistical test power  $(1-\beta)$  of 0.8, the sample size was determined to be 25 in each groups, by considering sample loss 100 eligible caregivers entered the study. Subjects were divided into three groups: the intervention group (33 members, spiritual group therapy), control group one (32 members, just group sessions without spiritual therapy) and control group two (35, neither spiritual group therapy nor group sessions), by block randomization. During the study, three members were excluded from the intervention group due to being absent more than twice, four from control group one due to absence or the death of the elderly under care, and four from control group two due to the death of the elderly under care or the failure to cooperate in completing the posttest questionnaire. Eventually, 89 participants were studied, including 30 members in the intervention group, 28 in control one and 31 in control two.

### RESEARCH DESIGN

The present experimental study was conducted in three groups on a population consisting of the main family caregivers of the elderly with Alzheimer's disease.

### MEASURES

Data was collected using a demographic information questionnaire and Robinson's (1983) Caregiver Strain Index. The demographic questionnaire consisted of two parts including the caregiver's personal information and the elderly's personal information. The section on the caregiver's personal information examined the variables of age, gender, education level, employment status, economic status, relation to the elderly and hours spent caring, and the section on the elderly's personal information examined the variables of age, gender, literacy status, insurance status, funding resources, length living with the caregiver and length afflicted with Alzheimer's.

Robinson's Caregiver Strain Index was produced by Robinson in 1983 to measuring care strain and consists of 13 items scored based on a four point Likert scale (very much = 4, a lot = 3, medium = 2 and little = 1) with a maximum score of 52 and a minimum score of 13. This tool has been used in studies conducted by Ilse, Feys, De Wit, Putman and De Weerd (2008) and Van Exel, Koopmanschap, van den Berg, Brouwer and Van den Bos (2005) (Ilse et al., 2008; Van Exel et al., 2005). In Robinson's study, the reliability of this tool was

<sup>1</sup> Human Immunodeficiency Virus.

calculated as 0.86 (Robinson, 1983). In Iran, the tool was used by Mohammadi and Babaie (2011) and Azimi, Mohammadi, Hosseini and Farzi (2013) on caregivers of elderly with Alzheimer's disease and caregivers of stroke patients, respectively. Both studies confirmed the reliability and validity of the tool. Tool reliability was calculated 0.86 by Cronbach's alpha and 0.74 in test retest (Azimi et al., 2013; Mohammadi & Babaie, 2011).

### ETHICAL CONSIDERATIONS

Ethical considerations according to the national and university ethical guidelines were respected including obtaining the necessary permissions from the university, presenting the project to Iran Alzheimer's Association and attracting their support, introducing the researcher to the study participants and explaining the study methods and objectives and obtaining informed written consent from the caregivers. The study participants were assured about the voluntary nature of participation, their right to withdraw from the study at any stage without any consequences for themselves or for their patients and the confidentiality of the data. They were told that the results would be forwarded to them, if they desired. The spiritual care intervention was performed for control groups one and two at the end of the study. Ethical certification was obtained from the university ethics committee, and the study was registered in the Iranian Registry of Clinical Trials on 27.12.2014 under the NO. IRCT2014111119898NI.

### PROCEDURE

The spiritual care training package was designed based on Richards and Bergin's (1997) spiritual model, which proposes a type of spiritual strategy. Interventions includes prayers for the caregivers and encouraging them to pray too, discussions about divine issues, using the holy books for treatment, using illustration and relaxation techniques, encouraging patients to forgive and sacrifice, helping patients to be in harmony with spiritual values, self-confessing spiritual beliefs and experiences, consulting with religious leaders and using religious book therapy (Hosseini, Salehi, Fallahi-Khoshknab, Rokofian, & Davidson, 2013). Content of the model was adjusted to the Iranian culture and Islam by religious experts and instructors of the University of Social Welfare and Rehabilitation Sciences. After obtaining permissions to study and use the registry from the university and Iran Alzheimer's Association and informed consent from the caregivers, the questionnaires were completed.

In the intervention group, Spiritual group therapy was held weekly for five weeks and each session lasted 45 to 60 min. Subjects presented included: Introduction and formation of group solidarity in session one, The role of reciting the Quran in personal peace in session two, The role of recommended recitals and repeat them in personal peace in session three, The role of reciting prayers in personal peace in session four and Explaining the experience of spiritual care and its effects, preparing

members to leave the group and take advantage of the group's achievements in session five.

Five, 45 to 60 minute weekly sessions were held for control group one. In these sessions caregivers were meet each other in groups, without performing any particular interventions, and daily issues were discussed. Control group two received no interventions, or even the chance of meeting the caregivers in the group. Demographic questionnaire and Robinson's Caregiver Strain Index were performed at the beginning of the study, and Caregiver Strain Index was completed again at the end of the study, after five weeks, in all three groups.

### STATISTICAL ANALYSIS

Descriptive statistics include mean  $\pm$  standard deviation and percentages were used to present samples demographic information, socio economic status and care strain scores. The Chi-squared test was used for categorical variables. In each group the difference between before and after mean of continuous variables was assessed by pair *t*-test. The comparison of means among three groups was done using one-way ANOVA followed by post hoc analysis. A *P* value of  $<0.05$  accepted as statistically significant. SPSS software package (SPSS Inc., Chicago) version 16 was applied for data analysis.

### RESULTS

Homogeneity tests confirmed that the three groups were statistically matched in terms of predisposing and confounding variables that could potentially affect the study results. Demographic data demonstrated that the majority of participants were female, married, housewives, had below secondary school education, average economic status, a parent child relation with the elderly under care and mean age was  $52.96 \pm 13.02$ . The majority of the elderly with Alzheimer's disease which cared by caregivers who participating in this study were female, illiterate, had insurance coverage, mean age was  $75.44 \pm 7.04$  and mean disease duration was  $6.07 \pm 3.49$  years. Table 1 illustrated that different education levels ( $P = 0.028$ ), economic status ( $P = 0.035$ ) and relationship with the elderly ( $P = 0.022$ ) had different care strain score. Fisher's least significant difference (LSD) indicated that care strain is higher in caregivers with education level of elementary and high school, low income and spouses.

In the intervention group, the mean score of care strain in pretest was  $37.16 \pm 1.26$  and  $32.43 \pm 2.73$  in posttest which decreased significantly ( $P < 0.001$ ). In control group one, the mean score of care strain in pretest was  $36.75 \pm 1.35$  and  $36.64 \pm 1.28$  in posttest which was not statistically significant ( $P = 0.375$ ). In control group two, the mean score of care strain was  $37.58 \pm 1.64$  in pretest and  $37.19 \pm 1.53$  in posttest which was not statistically significant ( $P = 0.831$ ); (Table 2).

Results showed that there were not statistically significant differences between the three groups in their mean pretest scores of care strain ( $P = 0.091$ ), but the posttest differences between the groups were significant ( $P < 0.001$ ). The post-hoc analysis showed that the

**Table 1**  
Socio economic status and pretest care strain scores in three groups.

| Socio-economic status         |                   | Mean $\pm$ SD    | N  | F    | P-value |
|-------------------------------|-------------------|------------------|----|------|---------|
| Education level               | Elementary school | $37.12 \pm 1.66$ | 32 | 3.18 | 0.028   |
|                               | High school       | $37.59 \pm 1.89$ | 37 |      |         |
|                               | College diploma   | $36.90 \pm 1.99$ | 10 |      |         |
|                               | University        | $36.10 \pm 1.59$ | 10 |      |         |
| Economic status               | Low               | $37.85 \pm 1.57$ | 7  | 3.49 | 0.035   |
|                               | Intermediate      | $37.31 \pm 1.34$ | 64 |      |         |
|                               | High              | $36.44 \pm 1.61$ | 18 |      |         |
| Relationship with the elderly | Child             | $37.31 \pm 1.36$ | 45 | 3.37 | 0.022   |
|                               | Spouse            | $37.43 \pm 1.41$ | 32 |      |         |
|                               | Grandchild        | $35.83 \pm 1.94$ | 6  |      |         |
|                               | In laws           | $38.16 \pm 1.98$ | 6  |      |         |

mean posttest score of care strain were significantly lower in the intervention group compared to the other two groups ( $P < 0.001$ ); however, no statistically significant differences were observed between control group one and control group two in their mean posttest scores of care strain ( $P = 0.154$ ); (Table 3).

## DISCUSSION

Current study examined the effect of spiritual group therapy on care strain in caregivers of the elderly with Alzheimer's disease in 2014. The results demonstrated that care strain reduced after the intervention. Because all three groups were demographically homogeneous, mean care strain change was due to spiritual care. Hence, the study hypothesis that spiritual care reduces care strain in home caregivers of the elderly with Alzheimer's disease was confirmed.

Care Strain was reported moderate to high in  $>50\%$  of caregivers of the elderly with Alzheimer's disease (Abdollahpour et al., 2012), and the majority of caregivers rated their problems severe or high (Zohari et al., 2006). Spurlock (2005) investigated spiritual health and its relationship with care strain in caregivers of patients with Alzheimer's disease, results showed that spiritual health has a significant role in caregivers life and can be a major coping mechanism for female caregivers (Spurlock, 2005). Spurlock study was a descriptive correlational study with a sample size of 150, consisting female caregivers, and used the Burden Interview Scale to assess care strain. Hence it cannot be compared to the present study. In mentioned study, spiritual care led spiritual health and a significant reduction in care strain occurred, so the results were similar to the current study.

Stuckey (2001) studied the role of religion and spirituality in the care of patients with Alzheimer's disease in a qualitative research, results illustrated that religious, spiritual beliefs and adherence to them can help to live stress free (Stuckey, 2001). Because stress is a major cause of care strain, these results are in consistent with the present study.

Spirituality is related to culture, religious and social status. Religious and culture are integrated in Iranian's lifestyle, most of Iranian people are Muslims. Religious beliefs play an important role in Iranian lives, especially in critical situations, and Religious is not separate from spirituality for Muslims (Asadi, 2013). Also, the meaning and interpretation of a critical experience are affected with personal life events and background (Boston, Mount, Orenstein, & Freedman, 2001), and they are differs in different societies.

Mohammadi and Babaee assessed the effect of participation in support groups on spiritual health and care strain in family caregivers of the elderly with Alzheimer in a single group of 32 caregivers. The results indicated that active participation in support groups empowered the caregivers for facing the difficulties of taking care of the elderly, and can help the caregiver to have effective coping with the complications of care by reducing care strain and promoting spiritual health (Mohammadi & Babaie, 2011). Type of intervention adopted by Mohammadi is different from the current study. Therefore these two studies cannot be compared, but they are similar in statistical population, care strain assessment tool and significant reduction in care strain.

**Table 3**

Between group comparison of pretest and posttest care strain scores in three groups.

| Groups   | Intervention |      | Control 1 |      | Control 2 |      | One way ANOVA |          |
|----------|--------------|------|-----------|------|-----------|------|---------------|----------|
|          | Mean         | SD   | Mean      | SD   | Mean      | SD   | F             | P-value  |
| Pretest  | 37.16        | 1.26 | 36.75     | 1.35 | 37.58     | 1.64 | 2.46          | 0.091    |
| Posttest | 32.43        | 2.73 | 36.64     | 1.28 | 37.19     | 1.53 | 57.57         | $<0.001$ |

Closer relationship between the caregiver and the patient causes more care strain (Andrén & Elmståhl, 2007). Caregiving of a spouse is difficult, because of the deterioration which is made in their spousal relationship (Savundranayagam, Montgomery, & Kosloski, 2010; Wennberg et al., 2015). Spousal caregivers spend more hours to provide care, and they are reluctant to receive help from other family members (Pinquart & Sörensen, 2006; Wennberg et al., 2015). Maybe these are the reason that spouses were suffered from the highest amount of care strain in the present study.

Financial strain is common in family caregivers of dementia patients as a result of providing long term care (Andrén & Elmståhl, 2007). Caregivers with higher financial level have more access to supportive services, so they bear lower care strain (Covinsky et al., 2003). Andrén and Elmståhl (2007) confirmed that lower income associated with higher level of care strain (Andrén & Elmståhl, 2007). In the present study, better economic status and higher education associated with lower level of care strain. The effect of education on care strain may because of the relationship between education and income.

Spiritual distress is common in caregivers of the elderly with Alzheimer's disease for many reasons, including fatigue, tribulation, sense of guilt, social isolation, economical hardships and so on. It seems that spiritual care can release negative emotions and increase coping abilities and ultimately affect care strain. The study limitations were the effect of caregiver's psychological status while completing the questionnaire, time limitations and lack of follow up and different severities of Alzheimer in the patients which may have affected the amount of care strain in the caregivers. Conducting a similar study with a longer follow up might be useful. By considering that a large number of elderly with Alzheimer's disease are kept in nursing homes, and professional caregivers are responsible for taking care of them, it is recommended to assess the effect of spiritual care on care strain in professional caregivers. Based on the results of the present study, Iran Alzheimer's Association is recommended to adopt spiritual care interventions along with the other care services and activities provided to caregivers. Because of the intense stress which families of patients with depression are experienced (Radfar, Ahmadi, & Fallahi-Khoshknab, 2014), it is recommended to assess the application spiritual care in this population in future research.

## CONCLUSION

Spiritual care can reduce care strain in home caregivers of the elderly with Alzheimer's disease. It can be used along with other care and supports and can facilitate the empowering of the home caregivers. Spiritual care is a simple, inexpensive and practical technique for caregivers of patients with Alzheimer's disease, and it can ease the burden of the disease. The application of the spiritual care intervention is recommended for closer relations and those with lower level of economic status with are more susceptible to the care strain.

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**Table 2**

Within group comparison of care strain scores before and after the interventions in each group.

| Groups       | Pretest |      | Posttest |      | Paired t-test |          |
|--------------|---------|------|----------|------|---------------|----------|
|              | Mean    | SD   | Mean     | SD   | T             | P-value  |
| Intervention | 37.16   | 1.26 | 32.43    | 2.73 | 13.08         | $<0.001$ |
| Control 1    | 36.75   | 1.35 | 36.64    | 1.28 | 0.902         | 0.375    |
| Control 2    | 37.58   | 1.64 | 37.19    | 1.53 | -0.215        | 0.831    |



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